

Understanding the Hurricane–Stock Market Dataset

A plain-English guide to every column in `complete_dataset.csv` — no finance background required.

This dataset pairs 420 hurricanes (2000–2025) with what the U.S. stock market was doing in the days before and after each storm. The goal is to explore a simple question: **do hurricanes move certain parts of the stock market in predictable ways?** For example, a hurricane barreling toward the Gulf of Mexico might spook energy-company investors before it even makes landfall, while insurance and construction companies might see effects only afterward, once the damage bill starts coming in.

Every hurricane in the dataset still carries its own storm details (name, date, category, deaths, damage, etc.) — this guide focuses on the **new columns added from the stock market**, since those are the ones that involve unfamiliar finance terminology.

The one-sentence version

For each hurricane, the dataset looks at four sectors of the stock market (Energy, Utilities, Financials, Industrials) and asks two things: **(1) how was that sector already behaving in the weeks before the storm?** and **(2) how did it perform in the five trading days afterward, compared to the market as a whole?** The first set of columns are the clues; the last four columns are the outcomes being explained.

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1. The five tickers behind every column

A **ticker** is just the short code a stock or fund trades under (like a username). The script doesn't track individual companies — it tracks five funds, each of which bundles many companies into one tradeable basket. That basket is called an **ETF (exchange-traded fund)**: think of it as a pre-made “grocery bag” of stocks that you can buy or sell as a single unit, so its price rises and falls with the group of companies inside it, not just one.

Ticker	What it actually is	Why it's in this dataset
SPY	Tracks the S&P 500 — roughly the 500 largest U.S. public companies.	Stands in for “the overall stock market.” Every sector's performance is measured <i>relative to</i> SPY, so we can tell whether a sector moved because of the hurricane specifically, or just because the whole market moved that day.
XLE	Energy sector fund — oil, gas, and refining companies.	Hurricanes routinely shut down Gulf of Mexico oil rigs and coastal refineries. This is the sector most likely to react <i>before</i> landfall, as a storm's forecast path becomes clear.
XLU	Utilities sector fund — electric and gas utility companies.	Utilities own the power lines and substations that hurricanes knock down. Storm damage means repair costs now, but also a captive customer base that has to keep paying — a useful contrast to Energy.
XLF	Financials sector fund — banks and, notably, insurance companies.	Insurers are on the hook for hurricane damage claims. A severe storm can mean a real hit to insurers' expected profits, which this fund should reflect.
XLI	Industrials sector fund — construction, machinery, and transport companies.	Rebuilding after a hurricane is Industrials' business: contractors, heavy equipment, building materials. This sector is the one most likely to benefit <i>after</i> a storm, once rebuilding demand becomes visible.
^VIX	Not a company at all — a market-wide “fear gauge.”	Explained in full in Section 2. Included because hurricanes, like any uncertain event, can make the whole market jumpier, separate from any one sector.

Why sectors instead of individual companies? A single company's stock price is noisy — it reacts to lawsuits, product launches, executive departures, and a hundred other things that have nothing to do with weather. Averaging together dozens of companies in the same industry cancels out most of that noise, so what's left over is more likely to be a genuine reaction to the hurricane.

2. Finance jargon, translated

These terms show up over and over in the column names. Once they click, the rest of the dataset reads much more easily.

Term	Plain-English definition
Trading day	A day the stock market is open (Monday–Friday, minus holidays). Markets are closed on weekends, so “5 trading days” usually spans about a full calendar week.
Close / closing price	The price of a stock or fund at the end of the trading day (4:00pm ET). This dataset uses closing prices throughout, since it's the standard, stable reference point used for day-to-day comparisons.
Return	The percent change in price over some stretch of time. A return of 0.02 means “up 2%.” A return of -0.05 means “down 5%.” All returns in this dataset are written as decimals, not percentages (0.02, not 2%).
Volatility	A measure of how much a price swings up and down, regardless of direction. High volatility doesn't mean a stock is going down — it means its price is jumpy and unpredictable either way. Calmer, more predictable prices have low volatility.
Volume	How many shares changed hands in a day. A spike in volume means a lot of investors suddenly decided to buy or sell — usually a sign that something newsworthy is happening.
Drawdown	How far a price has fallen from its recent high point, as a percentage. A drawdown of -0.10 means the price is currently 10% below its peak over the window being measured.
Beta	How sensitive a sector is to the market's overall moves, using SPY as the yardstick. A beta of 1.0 means the sector tends to move in step with the market. Above 1.0 means it swings <i>more</i> than the market (more aggressive); below 1.0 means it swings <i>less</i> (steadier). It does not measure whether a sector goes up or down — only how strongly it reacts.
Abnormal return	A sector's return <i>minus</i> the market's return over the same days. This is the key idea behind the whole dataset: if Energy is up 3% but the whole market is also up 3%, Energy didn't really do anything unusual — its abnormal return is 0. If Energy is up 3% while the market is flat, that 3% is genuinely abnormal, and plausibly hurricane-related.
VIX (the “fear index”)	A number calculated from options prices that reflects how much turbulence investors expect in the market over the next 30 days. It typically sits in the 12–20 range in calm periods and spikes above 30–40 during crises. It isn't a stock price and can't be “bought” directly — it's a mood ring for the whole market.

3. Timing columns — lining up the hurricane with a trading day

Hurricanes don't care what day of the week it is, but the stock market only exists on trading days. These two columns handle that mismatch.

Column	Plain-English meaning	Why it's in the dataset
<code>event_trading_date</code>	The trading day used to represent the hurricane in the market data.	If a hurricane's date falls on a weekend or holiday, the script rolls forward to the <i>next</i> day the market was actually open, since that's the earliest day the market could possibly react.
<code>event_date_shift_days</code>	How many calendar days that roll-forward took — 0 if the storm's date was already a trading day.	Lets you spot rows where the market's reaction is being measured a few days later than the storm itself, which matters if you're being strict about timing.

4. Pre-storm columns — what the market already knew

Every column in this section is calculated using **only trading days before** the hurricane's event day. These represent the information actually available to an investor at the moment the storm arrived — the conditions the hurricane hit into, not anything after. The pattern `{sector}_` means the same calculation is repeated separately for each of the four sectors (Energy, Utilities, Financials, Industrials).

Column	Plain-English meaning	Why it's in the dataset
<code>spy_prev_5d_return</code>	How much the overall market moved in the 5 trading days right before the storm.	A baseline mood check: was the market already rising, falling, or flat heading into the hurricane?
<code>vix</code>	The fear-index reading on the last trading day before the storm.	Tells you whether investors were already on edge (high VIX) or complacent (low VIX) when the hurricane arrived — context that can change how sharply the market reacts.
<code>{sector}_prev_5d_return</code>	That sector's own price return over the 5 trading days before the storm.	Sectors don't all start from the same place. A sector already sliding before a storm is a different situation than one that was already climbing.
<code>{sector}_20d_volatility</code>	How jumpy that sector's daily prices were over the prior 20 trading days, expressed on a yearly basis (annualized) so different time windows are comparable.	A sector that's already choppy may barely register a hurricane; a normally sleepy sector reacting sharply is more telling.

Column	Plain-English meaning	Why it's in the dataset
{sector}_volume_change	How the most recent day's trading volume compared to that sector's average volume over the previous 20 days.	A jump here means unusually heavy buying or selling was already underway right before the storm — a sign the market may have been anticipating it.
{sector}_drawdown	How far that sector was sitting below its own recent 20-day high, right before the storm.	Shows whether the sector was already under pressure (a large drawdown) or near its recent highs when the hurricane hit.
{sector}_beta	That sector's sensitivity to the overall market, measured over the prior 60 trading days.	Gives context for interpreting the sector's moves: a high-beta sector swinging hard might just be doing what it always does, not necessarily reacting to the storm.

5. Outcome columns — what actually happened afterward

These four columns are the payoff of the whole dataset — the thing all the pre-storm columns are trying to explain or predict.

Column	Plain-English meaning	Why it's in the dataset
<code>{sector}_5d_abnormal_return</code>	That sector's price return over the 5 trading days <i>after</i> the storm, minus the overall market's (SPY's) return over those same 5 days.	This isolates the hurricane-specific effect. A positive number means that sector outperformed the broader market in the week after the storm; a negative number means it lagged behind — in both cases, over and above whatever the market did generally.

Why subtract the market's return at all?

Suppose Industrials rose 4% in the five days after a hurricane. That sounds like a rebuilding boom — except if the entire stock market also rose 4% that week (say, because of unrelated good economic news), Industrials didn't actually outperform anything. Its *abnormal* return would be 0%. Subtracting the market's return is what lets you tell a real, storm-specific effect apart from the whole market simply having a good or bad week.

6. Why “before” and “after” are kept strictly separate

You may notice the dataset is careful to only use data from before the storm for the input columns in Section 4, and only data from after the storm for the outcome columns in Section 5, with the event day itself as the dividing line. This isn't just tidiness — it prevents a mistake called **look-ahead bias** (also called data leakage).

Imagine trying to predict whether Energy stocks will fall after a hurricane, but accidentally letting your prediction “peek” at what Energy stocks actually did in the days after the storm. The prediction would look perfect — because it's cheating, using information nobody actually had at the time. It's the equivalent of grading your own practice exam using the answer key sitting right next to it: you'd ace it, but learn nothing about how you'd do on the real test. Any model trained this way would look great in testing and then fail completely on a new, real hurricane, where the future genuinely isn't known yet.

That's why every pre-storm column is computed using a window that ends the day *before* the event trading day, and every outcome column begins the day *after* it.

7. Worked example — reading one full row

Here's the dataset's very first row, **Hurricane Alberto (2000)**, translated line by line. Alberto never made landfall, so it's a useful “quiet” example — a case where you'd expect the market reaction to be modest.

Field	Value	In plain English
hurricane_name	Alberto (2000)	The storm this row describes.
date	2000-08-03	The date recorded for the storm in the source hurricane data.
event_trading_date	2000-08-03	August 3, 2000 was itself a trading day (a Thursday), so no roll-forward was needed.
spy_prev_5d_return	-0.0054	The overall market had drifted down about half a percent in the week before Alberto formed — a quiet, near-flat market.
vix	20.0	A middling fear-index reading — not calm, not panicked.
energy_prev_5d_return	0.0238	Energy stocks were already up about 2.4% in the prior week, before the storm was even a factor.
energy_beta	0.19	A very low beta — over this particular 60-day window, Energy was barely moving in step with the broader market at all.
energy_5d_abnormal_return	0.0393	In the 5 trading days after Alberto's date, Energy outperformed the overall market by about 3.9 percentage points.
utilities_5d_abnormal_return	-0.0359	Utilities, by contrast, <i>underperformed</i> the market by about 3.6 points over the same days.

Putting it together: Alberto was a storm that stayed at sea and did essentially no damage, so on its own this row doesn't tell a dramatic story — and that's exactly the point of including quiet storms alongside destructive ones like Katrina or Ian. With hundreds of rows like this spanning calm storms and catastrophic ones, patterns worth trusting are the ones that show up consistently across many hurricanes — not the one-offs any single row happens to show.

8. Reading the numbers — what counts as big, small, positive, or negative

The worked example gave you real numbers, but not what to *do* with them. This section is a reference for both: what a positive vs. negative value means for each type of column, and roughly how large a value has to be before it's worth paying attention to.

“Significant” here means “worth a second look,” not a statistical test

The bands below are practical rules of thumb used to eyeball whether a number stands out — they are **not** a formal statistical significance test (like a p-value). With 420 hurricanes to compare, a rigorous analysis would test whether a pattern actually holds up across many storms before treating any single number as proof of a real effect. Treat these thresholds as a starting intuition, not a verdict.

Which way is which: reading the sign (+ / –)

Column	If negative	If positive
{sector}_prev_5d_return spy_prev_5d_return	Price fell over those 5 trading days.	Price rose over those 5 trading days.
{sector}_5d_abnormal_return	Sector underperformed the market by this many points over those 5 days.	Sector outperformed the market by this many points over those 5 days.
{sector}_volume_change	Trading was quieter than that sector's recent average.	Trading was busier than that sector's recent average.
{sector}_beta	Rare in this dataset — sector tends to move opposite the market.	Normal case — sector tends to move in the same direction as the market.

Three columns don't fit this +/- framing because they can't go both ways: `vix` and `{sector}_20d_volatility` are always zero or positive, and `{sector}_drawdown` is always zero or negative (you can only be at or below a recent peak, never above it). Their scales are covered below instead.

How big is big: magnitude thresholds

Read every row as: below the first number is noise, the middle range is worth noting, and past the last number is a genuinely large move. Values are shown as absolute size (ignore the sign, which you already read above).

Column	Noise	Notable	Large	Context
prev_5d_return	< 1%	1–3%	> 6%	A 5-day price move for a broad index or sector (applies to SPY and each sector alike). Under 1% is an ordinary week; past 6% is the kind of week that makes financial news.

Column	Noise	Notable	Large	Context
5d_abnormal_return	< 1 pt	1–2 pts	> 4–5 pts	The core “did this sector do something unusual?” number. Under 1 point, the sector basically just tracked the market. Past 4–5 points, that’s a real divergence unlikely to be pure chance.
vix	< 15	15–20	> 30	Below 15 is a calm, complacent market. 15–20 is normal. Above 30 signals real fear; above 40 is crisis-level (2008, March 2020).
20d_volatility	< 15%	15–25%	> 40%	Annualized price choppiness. Under 15% is calm; past 40% is crisis-level turbulence for that sector.
volume_change	< 20%	30–75%	> 100%	How much busier trading was vs. that sector’s own recent average. Past 100% means trading volume more than doubled — a clear spike in attention.
drawdown	< 3%	3–8%	> 15%	How far below its own recent peak the sector was sitting. Under 3% is essentially at its highs; past 15% is a serious decline.

Beta is a special case: it's not about direction at all

Beta value	What it suggests
0.3–0.7 (well below 1.0)	Defensive / muted — tends to move less than the market either way. Utilities often lands here.
≈ 1.0	Tends to move in step with the market — neither amplifying nor dampening its swings.
1.2–1.6+ (well above 1.0)	Amplifies the market’s moves — bigger swings in both directions. Financials and Industrials often run higher.
Near 0 or negative	Little to no relationship with the market, or an unusual inverse relationship. Uncommon — worth double-checking rather than assuming it’s meaningful.

Back to Alberto: applying these bands to the worked example

Revisiting Section 7’s numbers with these thresholds: Alberto’s `spy_prev_5d_return` of -0.54% is well under 1% — noise, confirming the market really was quiet beforehand. Its `vix` of 20.0 sits right at the edge between “normal” and “elevated,” not calm but not fearful either. Energy’s beta of 0.19 is well below the defensive 0.3–0.7 range — unusually muted for a sector that’s often more market-sensitive than that, at least across this particular 60-day window. And both abnormal returns — Energy at +3.9 points and Utilities at -3.6 points — land in the “real, noticeable divergence” range from the table above: too large to casually dismiss as noise, even for a hurricane that never made landfall.